

CS 302 - Formal Languages & Automata Theory Spring 2026

	<i>NAME / SCHEDULE</i>	<i>E-MAIL / PLACE</i>	<i>OFFICE HOUR</i>
<i>INSTRUCTOR</i>	<i>Kemal İNAN</i>	<i>inan</i>	<i>By appointment</i>
<i>TA</i>	<i>Alperen DOĞAN</i>	<i>alperend / FENS L045</i>	
<i>LA's</i>			
<i>LECTURES</i>	<i>Tu 13:40 - 14:30</i> <i>F 10:40 - 12:30</i>	<i>FENS L045</i> <i>FASS G018</i>	
<i>RECITATION</i>	<i>Tu 18.40 - 19.30</i>	<i>FASS 1010</i>	

Main Text: [*Introduction to Automata Theory, Languages and Computation*](#) , Hopcroft, Motwani & Ullman, Pearson (Addison Wesley) 2006 , 3rd edition

Auxiliary Text: *Elements of the Theory of Computation*, Lewis & Papadimitriou, Prentice Hall 1998.

Grading Policy: 35% Quizzes, 25% MT, 40% Final

Important: 10 quizzes with 15 or 20 minute durations each shall be part of the course. Quizzes will be held during chosen lectures. A student missing more than **3** quizzes **fails** irrespective of medical or any other excuse! The overall quiz grading will be the average of the best **7** out of **10**.

Homework Policy: There will be 8 HWs as part of the course. HWs will NOT be collected or graded ; but questions throughout all exams and quizzes may involve HW topics and questions.

Tentative Schedule (green dates = Friday 2 hour lectures)

<i>February</i>					17 1	20 1,2 <i>HW1</i>	24 2	27 2,3
<i>March</i>	3 3 <i>HW2</i>	6 3,4	10 4	13 5	17 5 <i>HW3</i>	24 6	27 6,8	31 8
<i>April</i>	3 8,7 <i>HW4</i>	7 7	10 7	14 <i>Midterm</i>	17 9	21 10,11	24 12 <i>HW5</i>	28 12
<i>May</i>	5 12	8 12,13 <i>HW6</i>	12 14	15 15	19 16	22 16 <i>HW7</i>		

Tentative Course Outline

- 1 – Introduction: Languages, Automata and Grammars (Main Text (MT) 1.1, 1.5) Slide1***
- 2 – Deterministic Finite Automata as Language Acceptors (DFA) (MT 2.1, 2.2) S2***
- 3 – Nondeterministic Finite Automata (NFA) and Linguistic Equivalence to DFA (MT 2.3 - 2.5) S2***
- 4 – Regular Expressions (RE) (MT 3.1) S3***
- 5 – RE and NFA (M.T 3.2) S3***
- 6 – Regular Languages and Properties (M.T. 4.1, 4.2) S3-S4***
- 7 – State Equivalence and Minimal State DFA (MT 4.4) S4***
- 8 – Algorithms for the DFA and NFA (MT 4.3, 4.4) S4***
- 9 – Context-Free Grammars (CFG) (MT 5.1) S5***
- 10– Parse Trees and Applications (MT 5.2, 5.3) S5***
- 11 – Ambiguity in Grammars and Languages (M.T. 5.4) S5***
- 12 – Pushdown Automata (PDA) (MT 6.1, 6.2) S6***
- 13 – CFG and PDA (M.T. 6.3) S6***
- 14 – Deterministic Context-Free Languages (MT 6.4) S6-S7***
- 15 – Properties of and Algorithms for Context Free Languages (MT 7.1,7.2, 7.4) S7***
- 16 - Determinism and Parsing (AT 3.7, p 158-177) S7***
- 17 - Introduction to Turing Machines (AT Chapters 4 ,5 selections) S8***